

SHIYAO WANG

Senior Software Engineer

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LINKS

[*LinkedIn*](#)

PROFILE

Frontend-leaning Full Stack Senior Engineer with 8+ years of experience at Netflix, Amazon, and Spotify. Expert in **React, Python, and Java**, delivering **scalable, user-focused applications** that power millions of users. Proven track record of leading **frontend modernization, ad-tech platforms, and super-app features** with strong cross-functional collaboration. Passionate about building **high-performance interfaces** that drive measurable business impact.

SKILLS

❖ Frontend Development

React, React Native, JavaScript (ES6+), TypeScript, HTML5, CSS3, WebSockets, Server-Side Rendering (SSR), Responsive Design, Accessibility (WCAG)

❖ Backend & Data

Python, Java (Spring Boot), Node.js, C++, C, REST APIs, GraphQL, MySQL, PostgreSQL, MongoDB, Redis, Hadoop, Apache Spark, Kafka, Event-driven Architecture

❖ Cloud & Infrastructure

AWS (Lambda, DynamoDB, S3, CloudFront, SNS/SQS), Docker, Kubernetes, Microservices Architecture, CI/CD (Jenkins, GitHub Actions), CDN Integration, Feature Flags

❖ Testing & Engineering Practices

Jest, Mocha, Cypress, Unit & Integration Testing, A/B Testing Platforms, Performance Optimization, Secure Development (HIPAA-compliant APIs), Agile/Scrum Collaboration

EMPLOYMENT HISTORY

❖ Senior Software Engineer, Netflix

Dec 2023 — Present
Remote

- **Spearheaded** the design and development of Netflix's in-house ad-serving platform by leveraging **Java (Spring Boot), React, GraphQL, Python ML pipelines, and CDN integration**, enabling seamless, high-quality ad delivery for over **270M global subscribers**.
- **Architected** a low-latency **Ad Decision Service** with **Java microservices, in-memory caching, and ML-based pacing algorithms**, achieving ad decision times under **50ms** and ensuring balanced campaign delivery across billions of impressions.
- **Engineered** client-server integration for ad playback by building **REST/JSON APIs, feature-flag rollouts, and SSAI/CSAI hybrid strategies**, reducing ad startup buffering by **30%** and maintaining Netflix's high playback standards.
- **Developed** internal **React/D3 dashboards and creative review tools** to streamline ad operations workflows, cutting manual review and monitoring tasks by **40%** and accelerating ad campaign launches.

- **Optimized** frontend performance with **React SSR, code-splitting, and CDN edge caching**, achieving faster homepage load times and supporting concurrent A/B tests across multiple markets.
- **Orchestrated** cross-functional collaboration with ML researchers, client engineers, CDN experts, and ad ops teams to implement **episodic campaigns and contextual targeting**, expanding Netflix's differentiated ad formats.
- **Championed** a culture of ownership and innovation by **authoring design docs, mentoring junior engineers, and driving post-mortem reviews**, reinforcing team agility and resilience in a high-growth product area.

❖ Senior Software Engineer, Amazon May 2022 — Dec 2023 Remote

- **Led** modernization of Amazon's **Demand-Side Platform (DSP)** by developing a **React/TypeScript frontend with GraphQL APIs and Java microservices**, simplifying campaign creation for thousands of advertisers and scaling to **billions in ad spend**.
- **Built** a step-by-step **Campaign Builder wizard** using **React Router, Redux, and dynamic imports**, reducing campaign setup time by **25%** and improving usability scores in advertiser surveys.
- **Implemented** GraphQL data aggregation and **Redis-based caching layers** across campaign performance dashboards, cutting average data retrieval latency from **800ms to under 200ms**.
- **Integrated** event-driven workflows with **Amazon's internal event bus, SNS/SQS, and idempotent API design**, ensuring reliable cross-service communication for billing, targeting, and ad-serving systems.
- **Enhanced** system reliability through **feature flags, canary deployments, load testing, and chaos simulations**, reducing incident recovery time by **35%** while supporting continuous delivery pipelines.
- **Cultivated** strong cross-functional alignment by collaborating with **designers, SREs, and data scientists**, facilitating smoother feature rollouts and reinforcing Amazon's customer-obsessed culture.

❖ Software Engineer , Spotify Jan 2020 — May 2022 Remote

- **Directed** frontend development of Spotify's **Ads Manager web app** using **React, TypeScript, Redux, GraphQL, and code-splitting**, improving advertiser workflows and supporting global campaign launches.
- **Streamlined** backend integrations by **building Python/Django services, Redis caches, and Kafka pipelines** for ad targeting and scheduling, supporting millions of ad requests per second with sub-200ms latency.
- **Implemented** A/B testing instrumentation through **Spotify's experimentation platform (Confidence), analytics event logging, and feature-flagged rollouts**, enabling data-driven validation of interactive ad formats.
- **Optimized** advertiser-facing dashboards with **lazy loading, virtualized lists, and GraphQL schema design**, reducing interface load times by **40%** for advertisers with large campaign portfolios.
- **Collaborated** across squads with **designers, mobile engineers, and SREs**, fostering knowledge-sharing and agile delivery practices that boosted overall team velocity and product innovation.

❖ Software Engineer, Grab Oct 2018 — Jan 2020 Remote

- **Pioneered** adoption of **React Native** for Grab's GrabPay and super-app modules, delivering shared codebases across iOS and Android, reducing development effort by **50%** for multi-platform releases.
- **Integrated** React Native modules into native codebases using **Java/Kotlin bridges, Swift/Obj-C bindings, and Node.js API layers**, enabling seamless hybrid app experiences at super-app scale.
- **Accelerated** feature delivery with **hot-reloading, shared UI libraries, and TypeScript adoption**, cutting iteration cycles from weeks to days for payment and transaction flows.
- **Partnered** with distributed engineering teams across Southeast Asia to define **React Native integration guidelines and internationalization strategies**, ensuring consistent multi-language user experiences.

EDUCATION

❖ University of Iowa 2016 — 2018 *M.S. in Computer Science*

❖ Denison University 2012 — 2016 *B.S. in Computer Science*